Assistant Professor Exam – 2024

ECONOMICS

UNIT – 2 : MACRO-ECONOMIC ANALYSIS

Detailed Notes with Diagrams (Hinglish)

Covers every topic of the official syllabus:

National Income – Definition, Concepts, Measurement, Social Accounting & Welfare

Determination of Output & Employment – Classical vs Keynesian Approach

Keynesian Employment Theory; Aggregate Supply & Demand; Effective Demand

Liquidity Preference Theory

Psychological Law of Consumption; Absolute, Relative, Permanent & Life-Cycle Income Hypotheses

Monetary Policy – Instruments & Development of Monetary Policy in India

Multiplier Theory; Theories of Investment & Accelerator; Four-Sector Output–Price Determination

Fleming–Mundell Open Economy Model; IS–LM Model; Trade Cycle Theories

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1. National Income: Concepts, Measurement & Social Accounting

1.1 Definition & Meaning

Macro Economics economy ko ek **whole** ke roop mein study karti hai — aggregate variables jaise total output, total employment, general price level, inflation aur growth. Iska sabse central concept hai **National Income**.

National Income = ek desh ke **normal residents** dwara ek **accounting year** (usually 1 April – 31 March) mein produce ki gayi sabhi final goods & services ki **money value**. Marshall, Pigou aur Fisher ne alag-alag definitions di:

- **Marshall (Production angle):** ek desh ke labour aur capital se produce hone wali net aggregate of commodities.
- **Pigou (Income angle):** wo part of national dividend jo money mein measure kiya ja sake.
- **Fisher (Consumption angle):** ek saal mein consume hone wali goods & services se milne wali real income/services.

1.2 Important Concepts (Aggregates)

National income ke kai related measures hain — inhe samajhna zaruri hai:

Concept	Meaning / Formula
GDP (Gross Domestic Product)	Desh ki domestic boundary ke andar produced final goods/services ki value (1 year).
GNP (Gross National Product)	GDP + NFIA (Net Factor Income from Abroad). National residents ki.
NDP (Net Domestic Product)	GDP − Depreciation (consumption of fixed capital).
NNP (Net National Product)	GNP − Depreciation.
NNP at Factor Cost = National Income	NNP at Market Price − Indirect Taxes + Subsidies.
Personal Income (PI)	Households ko actually receive hone wali income.
Disposable Income (DI)	PI − Direct (personal) taxes. = Consumption + Saving.
Per Capita Income	National Income ÷ Total Population.

GNP = GDP + NFIA
NNP (at FC) = GNP − Depreciation − Net Indirect Taxes
National Income = NNP at Factor Cost

Market Price vs Factor Cost

Market Price (MP) mein indirect taxes (jaise GST) included hote hain aur subsidies nahi. **Factor Cost (FC)** actual factors ko mili payment hai. Relation: **FC = MP − Indirect Taxes + Subsidies**. Net Indirect Tax (NIT) = Indirect Tax − Subsidy.

1.3 Methods of Measurement

National income measure karne ke teen methods hain — teeno theoretically same value dete hain (circular flow ke teen phases):

1. **Product / Value-Added Method:** har producing unit ki 'value added' (output value − intermediate consumption) jodte hain → GDP at MP. Double counting se bachne ke liye sirf 'value added' ya 'final goods' lete hain.
2. **Income Method:** sabhi factors ki income jodte hain — Compensation of Employees (wage), Rent, Interest, Profit + Mixed income = NDP at FC.
3. **Expenditure Method:** final expenditure jodte hain → GDP at MP =

$$\text{GDP (MP)} = C + I + G + (X - M)$$

(C = consumption, I = investment, G = government spending, X–M = net exports)

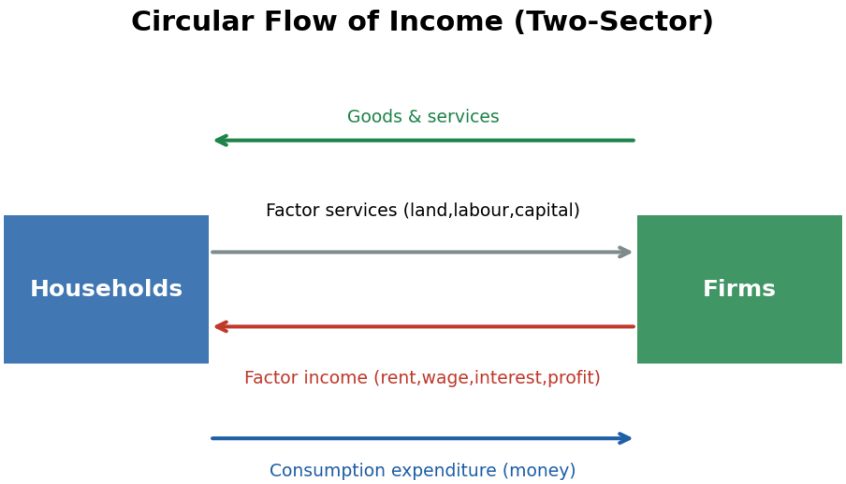


Fig 1.1 — Circular Flow of Income: real flows (goods, factor services) aur money flows (income, spending) opposite directions mein.

Precautions in Measurement

(i) Transfer payments (pension, scholarship, gifts) include **nahi** karte (koi production nahi). (ii) Second-hand goods ki sale exclude. (iii) Sirf **final goods** — intermediate goods nahi (double counting). (iv) Illegal/black income aur barter/self-consumption ka measurement mushkil. (v) Owner-occupied house ka 'imputed rent' include karte hain.

1.4 National Income & Social Accounting

Social Accounting (Richard Stone) ek systematic framework hai jismein poori economy ke transactions ko **sectors** (households, firms, government, rest of the world) aur accounts mein record kiya jaata hai. Isse **input-output tables, flow of funds, balance of payments aur national income accounts** banaye jaate hain. India mein ye kaam **CSO/NSO (MoSPI)** karta hai; RBI bhi flow-of-funds publish karta hai.

1.5 National Income & Welfare

National income ko aksar **welfare ka indicator** maana jaata hai (zyada income = zyada welfare), par ye perfect measure **nahi** hai. Limitations:

- **Distribution ignore:** Per capita income badh sakti hai par inequality bhi badh sakti hai.
- **Non-market activities:** Housewife ka kaam, self-consumption count nahi hote.
- **Externalities / pollution:** Production se environmental damage GDP se ghata nahi jaata (rather repair spending GDP badha deti hai).
- **Quality of life / leisure:** Health, education, leisure jaise factors capture nahi hote. Isi liye HDI, Green GDP, NEW (Net Economic Welfare) jaise measures suggest kiye gaye.

2. Determination of Output & Employment

2.1 Classical Approach

Classical economists (Adam Smith, Ricardo, J.B. Say, Pigou, Marshall) maante the ki economy mein hamesha **full employment** rehta hai (long run), aur unemployment sirf temporary/voluntary hota hai. Iske 3 pillars:

1. **Say's Law of Markets:** *"Supply creates its own demand."* Production ke process mein hi utni income generate ho jaati hai jisse poora output khareed liya jaata hai — isliye general overproduction (glut) possible nahi.
2. **Wage-Price Flexibility:** Labour market mein real wage flexible hone se demand=supply ho jaata hai → full employment automatically restore hota hai.
3. **Saving = Investment via interest rate:** Flexible interest rate savings aur investment ko equal kar deta hai, isliye demand kabhi kam nahi padti.

Classical system mein output aur employment **labour market** mein determine hote hain (real wage = MPL), aur **money neutral** hota hai (money sirf price level decide karta hai — Quantity Theory of Money).

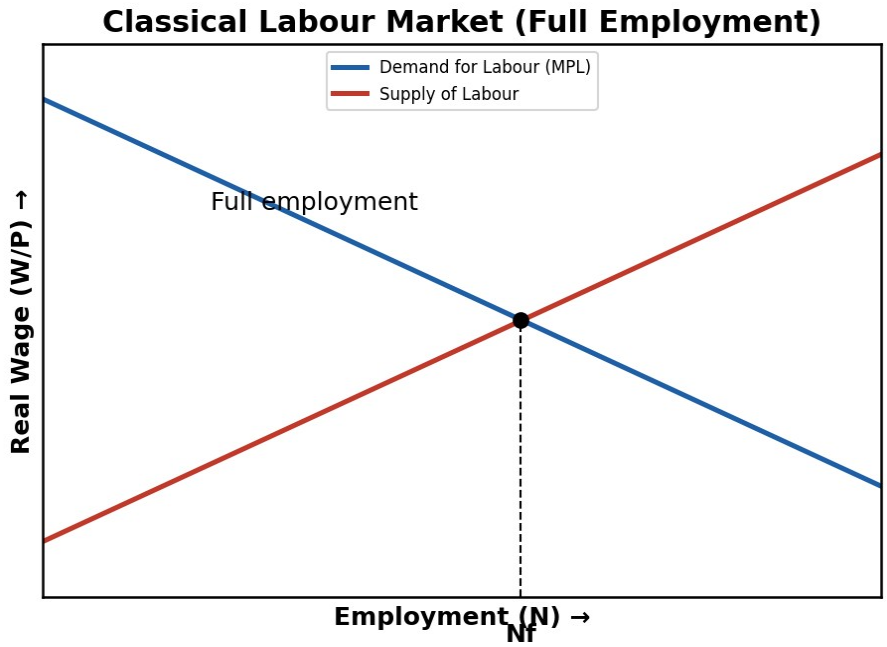


Fig 2.1 — Classical labour market: demand (MPL) = supply par full employment Nf; wage flexible.

Keynes ki Classical par Criticism

Keynes ne 1936 mein *"General Theory of Employment, Interest and Money"* mein dikhaya ki (i) Say's Law galat hai — demand apne aap supply nahi banati; (ii) wages downward **rigid** hoti hain; (iii) economy **under-employment equilibrium** mein bhi reh sakti hai (Great Depression isi ka example). Full employment **special case** hai, general nahi.

2.2 Keynesian Approach

Keynes ke according output aur employment **Aggregate Demand (AD)** se determine hote hain (short run, prices/wages assumed sticky). Jitni demand, utna output aur employment. Central concept hai **Effective Demand**.

Aggregate Demand (ADF) & Aggregate Supply (ASF)

- **Aggregate Supply Price (ASF):** wo minimum total receipts jo producers ko different employment levels par production karne ke liye chahiye — employment ke saath badhti hai.
- **Aggregate Demand Price (ADF):** wo total receipts jo producers ko different employment levels par milne ki expectation hai (= total expected spending C + I).

Point of Effective Demand

Effective Demand wo point hai jahan **ADF = ASF** — yani jahan entrepreneurs ki expected receipts aur required receipts barabar ho jaayein. Yahi level of **employment & output** determine karta hai. Important: ye level **full employment se kam** bhi ho sakta hai → 'under-employment equilibrium'.

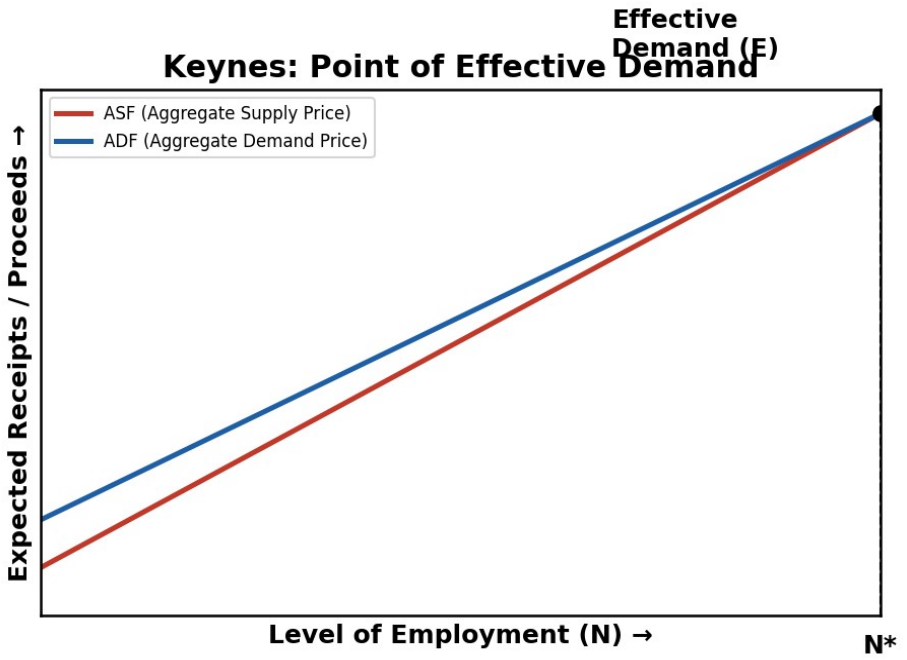


Fig 2.2 — ADF aur ASF ka intersection 'Effective Demand' point E; yahan employment N tay hota hai.*

Components of Aggregate Demand

AD = **C + I + G + (X − M)**. Keynes ne kaha consumption (C) income ke saath stable badhti hai, par **investment (I)** volatile hota hai (depends on MEC & interest rate) — isliye AD mein fluctuation mostly investment se aati hai.

3. Keynesian Income Determination & AS–AD

3.1 Two Approaches to Equilibrium Income

Keynesian model mein equilibrium income do equivalent tareeke se nikalti hai:

(a) Aggregate Demand = Aggregate Supply (Income–Expenditure / 45° approach)

Equilibrium wahan jahan **planned Aggregate Expenditure (AE) = Output (Y)** — graph mein AE curve **45° line** ko cut karti hai. 45° line par har point Y = AE hota hai.

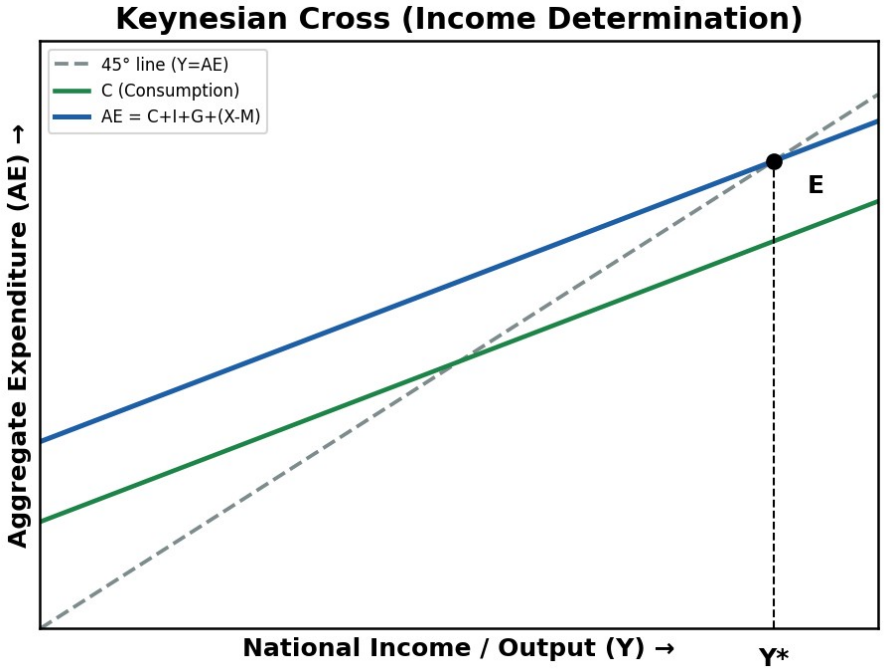


Fig 3.1 — Keynesian Cross: AE = C+I+G+(X–M) jahan 45° line ko cut kare wahi Y (equilibrium income).*

(b) Saving = Investment approach

Equilibrium par **planned S = planned I**. Agar S > I → demand kam → output ghatega; agar I > S → demand zyada → output badhega; balance par equilibrium.

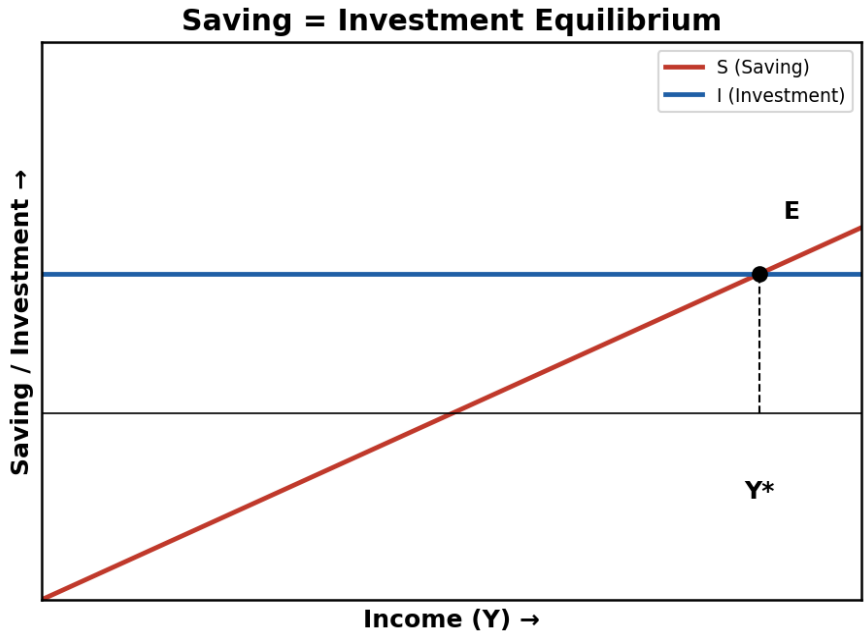


Fig 3.2 — S aur I curves ka intersection equilibrium income Y deta hai (Keynesian).*

3.2 Aggregate Demand – Aggregate Supply (Price level included)

Jab price level ko bhi variable maante hain to **AD–AS model** use hota hai. **AD curve** downward sloping (price ↓ → real demand ↑), **AS curve** upward (Keynesian range mein flat, classical range mein vertical). Inka intersection **equilibrium output aur price level** dono determine karta hai.

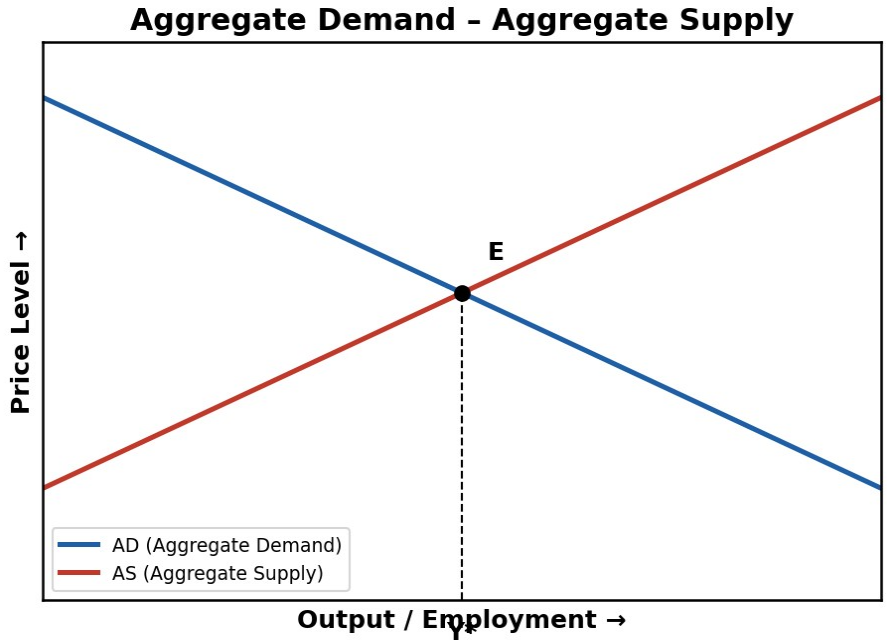


Fig 3.3 — AD–AS intersection se equilibrium output (Y) aur general price level tay.*

4. Consumption Function & Income Hypotheses

4.1 Keynes's Psychological Law of Consumption

Keynes ka **Fundamental Psychological Law:** *"As income increases, consumption increases but by less than the increase in income."* Yani income badhne par consumption badhta hai par income se kam ratio mein — baaki income save ho jaati hai.

Consumption function: C = a + bY, jahan a = autonomous consumption (income 0 par bhi zaruri kharch), b = MPC.

$$C = a + bY$$
$$APC = C / Y \qquad MPC = \Delta C / \Delta Y$$
$$MPC + MPS = 1$$

- **MPC (Marginal Propensity to Consume):** extra income ka jo hissa consume hota hai (ΔC/ΔY); 0 < MPC < 1.
- **APC (Average Propensity to Consume):** total consumption ÷ total income (C/Y); income badhne par APC girti hai.
- **MPS / APS:** saving wale counterparts; MPS = 1 − MPC.

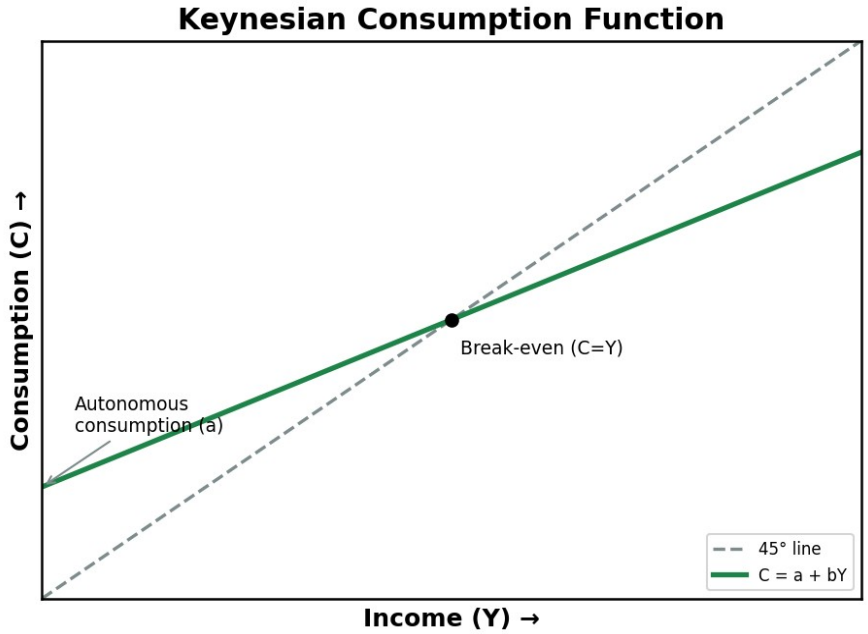


Fig 4.1 — Consumption function C=a+bY; 45° line ke saath cut point = break-even (C=Y).

Three Propositions of the Law

(i) Jab income badhti hai, consumption **badhta** hai par kam ratio mein. (ii) Badhi hui income consumption aur saving **dono** mein bant-ti hai. (iii) Income badhne par **dono** (consumption & saving) badhte hain — yani APC girti hai aur APS badhti hai.

4.2 Theories of Consumption (Income Hypotheses)

Keynes ke 'absolute income' idea ke baad teen aur famous hypotheses aaye jo consumption ko behtar explain karte hain:

(A) Absolute Income Hypothesis (Keynes)

Consumption sirf **current absolute (disposable) income** par depend karta hai. Income badhe to APC gire. **Problem (puzzle):** short-run data mein APC girti dikhi, par **Kuznets** ke long-run data mein APC **constant** nikli — yani Keynes ki theory long run ko explain nahi kar payi. Isi ko solve karne aage ki theories aayi.

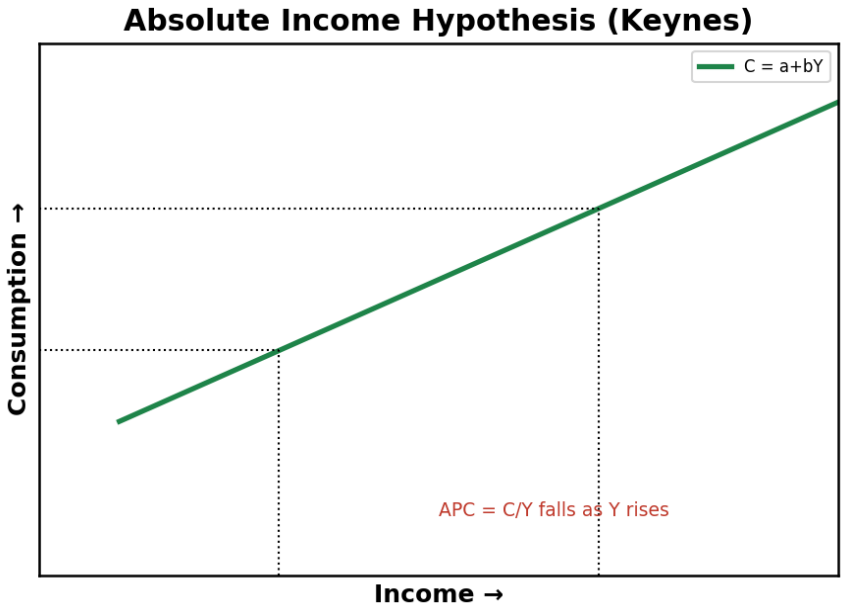


Fig 4.2 — Absolute Income Hypothesis: income badhne par APC (C/Y) girti hai.

(B) Relative Income Hypothesis (Duesenberry)

Consumption **absolute income par nahi, balki 'relative' income** par depend karta hai — yani doosron ke comparison mein aur apne hi past peak income ke comparison mein. Do effects:

- **Demonstration Effect:** log apne se ameer logon ko dekh kar consume karte hain (keeping up with the Joneses).
- **Ratchet Effect:** income girne par log apni consumption easily kam nahi karte (past peak income ki aadat) — consumption 'ratchet' ki tarah upar atak jaati hai.

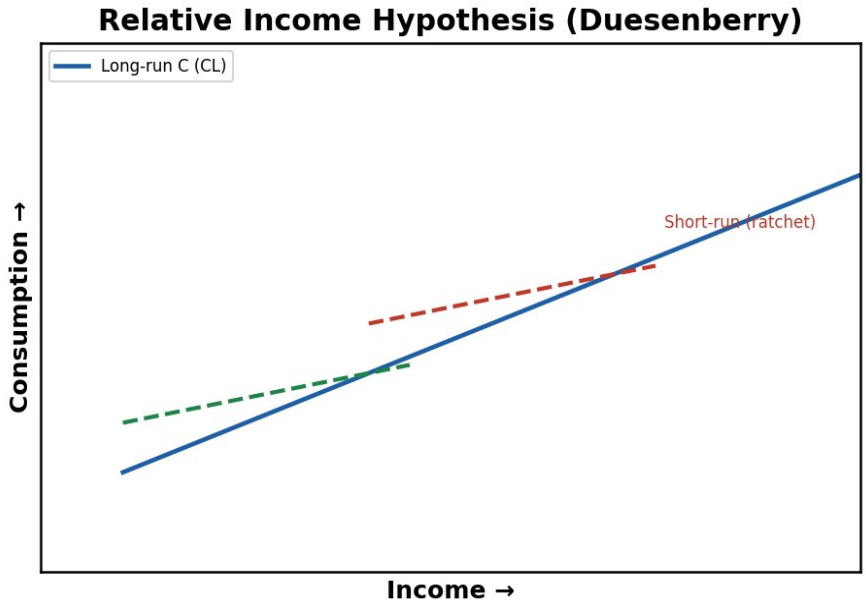


Fig 4.3 — Relative Income: short-run consumption 'ratchet' karti hai; long-run line CL steeper.

(C) Permanent Income Hypothesis (Friedman)

Milton Friedman: consumption **permanent income (Yp)** par depend karta hai, current measured income par nahi. Measured income = Permanent + Transitory; consumption bhi permanent + transitory.

$$C_p = k \cdot Y_p \quad (k = \text{constant proportion})$$

Transitory income (jaise bonus, lottery) ko log mostly **save** kar dete hain, consume nahi. Isliye long run mein APC stable rehti hai. Ye Kuznets puzzle solve karta hai.

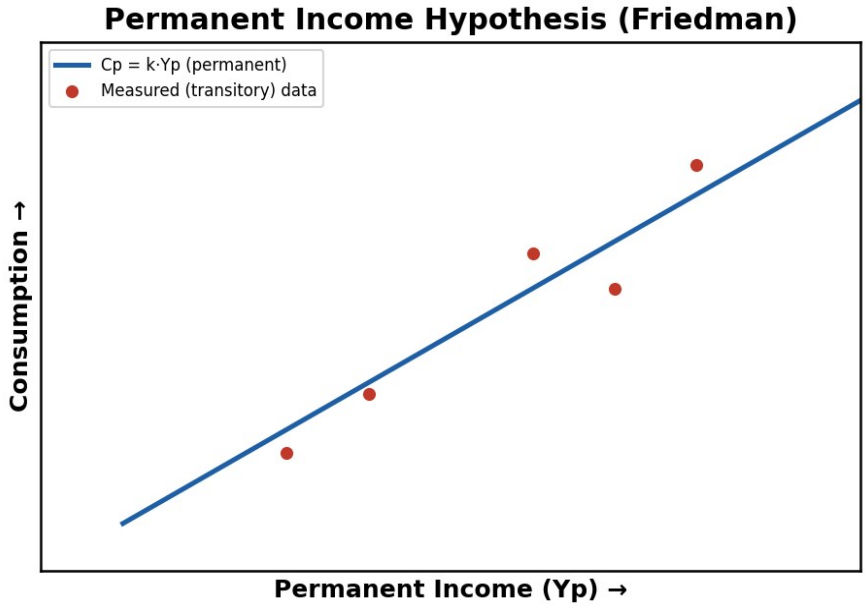


Fig 4.4 — Permanent Income: consumption permanent income ke proportion mein; transitory shocks scatter create karte hain.

(D) Life-Cycle Hypothesis (Modigliani, Ando, Brumberg)

Log apni **poori zindagi (life-cycle)** ki expected income ko dhyaan mein rakh kar consumption ko **smooth** karte hain. Young age mein income kam → borrow/dissave; middle age mein income high → save; old age (retirement) mein → dissave (savings kharch).

Implication: Consumption current income se zyada **lifetime resources (wealth + income)** par depend karta hai — isliye short-run aur long-run consumption behaviour dono explain ho jaate hain.

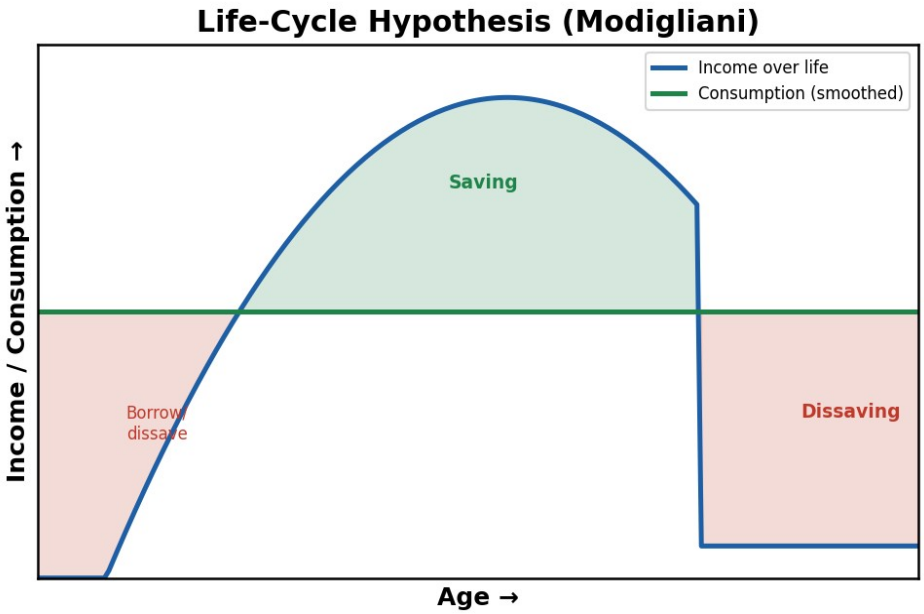


Fig 4.5 — Life-Cycle: consumption almost constant; income hump-shaped → young & old mein dissaving, middle age mein saving.

Four Hypotheses — One-line Recall

Absolute (Keynes): current income. **Relative (Duesenberry):** relative income — demonstration + ratchet effect. **Permanent (Friedman):** permanent income; transitory income saved. **Life-Cycle (Modigliani):** lifetime resources; consumption smoothing across age.

5. Monetary Policy

5.1 Meaning & Objectives

Monetary Policy central bank (India mein **RBI**) dwara **money supply** aur **interest rates** ko control karke economic objectives achieve karne ki policy hai. Main objectives: price stability (controlling inflation), economic growth, full employment, exchange rate stability aur financial stability. India mein 2016 se **Flexible Inflation Targeting (FIT)** framework hai — target **4% CPI inflation (±2%)**, jo **MPC (Monetary Policy Committee)** decide karti hai.

5.2 Instruments of Monetary Policy

RBI ke instruments do tarah ke hote hain — quantitative (general) aur qualitative (selective):

(A) Quantitative / General Instruments

- **Repo Rate:** wo rate jis par RBI banks ko short-term loan deta hai. Repo ↑ → borrowing mehengi → money supply ↓ (anti-inflation).
- **Reverse Repo Rate:** wo rate jis par RBI banks se money absorb karta hai.
- **Bank Rate:** long-term lending rate (penal rate ke roop mein bhi).
- **CRR (Cash Reserve Ratio):** deposits ka jo % banks ko cash mein RBI ke paas rakhna hota hai. CRR ↑ → lending capacity ↓.
- **SLR (Statutory Liquidity Ratio):** deposits ka jo % banks ko liquid assets (govt securities, gold) mein rakhna hota hai.
- **Open Market Operations (OMO):** RBI ka govt securities ko buy/sell karna. Sell → money soak; Buy → money inject.
- **MSF (Marginal Standing Facility) & LAF (Liquidity Adjustment Facility):** daily liquidity manage karne ke tools.

(B) Qualitative / Selective Instruments

- **Margin Requirements:** loan ke against collateral ka minimum margin.
- **Credit Rationing:** kuch sectors ke liye credit limit.
- **Moral Suasion:** RBI ki banks se request/persuasion.
- **Direct Action:** rules todne par penalties.

Situation	Policy Type	RBI Action
Inflation (zyada money)	Tight / Dear Money	Repo ↑, CRR ↑, SLR ↑, OMO Sell
Recession (kam demand)	Easy / Cheap Money	Repo ↓, CRR ↓, SLR ↓, OMO Buy

5.3 Development of Monetary Policy in India

1. **1935–1950s:** RBI establishment (1935); early focus credit regulation aur WWII finance par.
2. **1950s–1980s:** Planned development era — 'credit planning', priority sector lending, controlled interest rates; deficit financing ke saath inflation pressure.
3. **1985–1990:** Chakravarty Committee (1985) ne monetary targeting suggest kiya — money supply (M3) growth ko control karna.
4. **1991 ke baad (Reforms):** Liberalisation; interest rates deregulate, CRR/SLR gradually kam, market-based instruments (OMO, LAF 2000) introduce.

5. **2014–2016:** Urjit Patel Committee → 'Flexible Inflation Targeting'; CPI ko nominal anchor banaya; 2016 mein statutory 'MPC' (6 members) ka gathan.
6. **Present:** 4% (±2%) CPI inflation target; repo rate primary instrument; transparency aur communication (bi-monthly policy) par focus.

6. Multiplier, Investment, Accelerator & Four-Sector Model

6.1 Investment Multiplier (Keynes)

Multiplier (k) batata hai ki autonomous investment (ya spending) mein ek change se national income mein **kitna guna** change aata hai. Kyunki ek vyakti ka kharch doosre ki income banta hai, spending rounds mein circulate hoti hai.

k = ΔY / ΔI = 1 / (1 − MPC) = 1 / MPS
Example: MPC=0.8 → k = 1/(1−0.8) = 5

Multiplier ki value MPC par depend karti hai — MPC zyada → multiplier zyada. Multiplier **leakages** (saving, taxes, imports) se kam ho jaata hai.

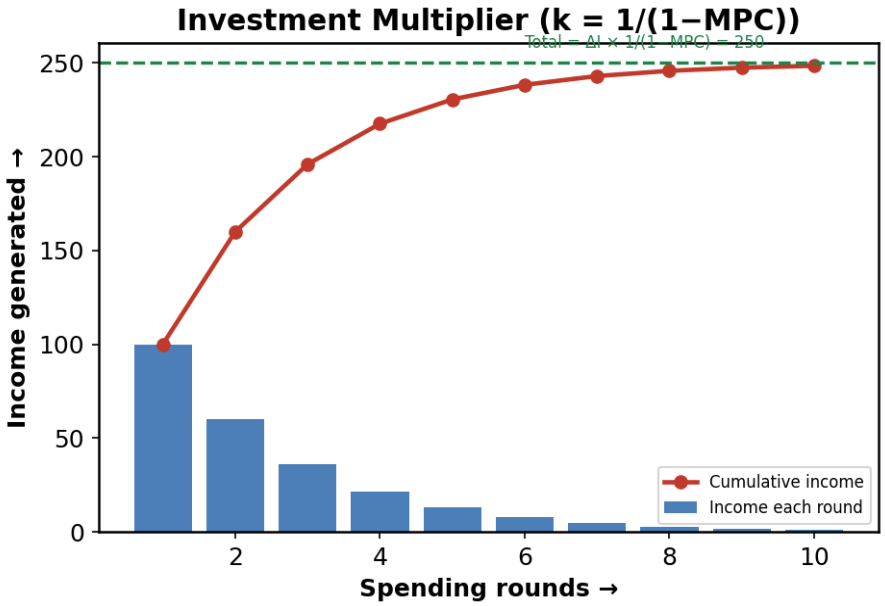


Fig 6.1 — Multiplier process: har round mein income (MPC times) ghatti jaati hai; total income = ΔI × 1/(1−MPC).

6.2 Theories of Investment

Investment do tarah ka: **Autonomous** (income se independent — govt, infrastructure) aur **Induced** (income/demand badhne se). Keynes ke according investment do cheezon par depend karta hai:

- **Marginal Efficiency of Capital (MEC):** ek extra unit capital se expected rate of return.
- **Rate of Interest (r):** borrowing ki cost.

Firm tab tak invest karti hai jab tak **MEC > r**; equilibrium investment wahan jahan **MEC = r**. MEC ↓ ya r ↑ → investment ↓.

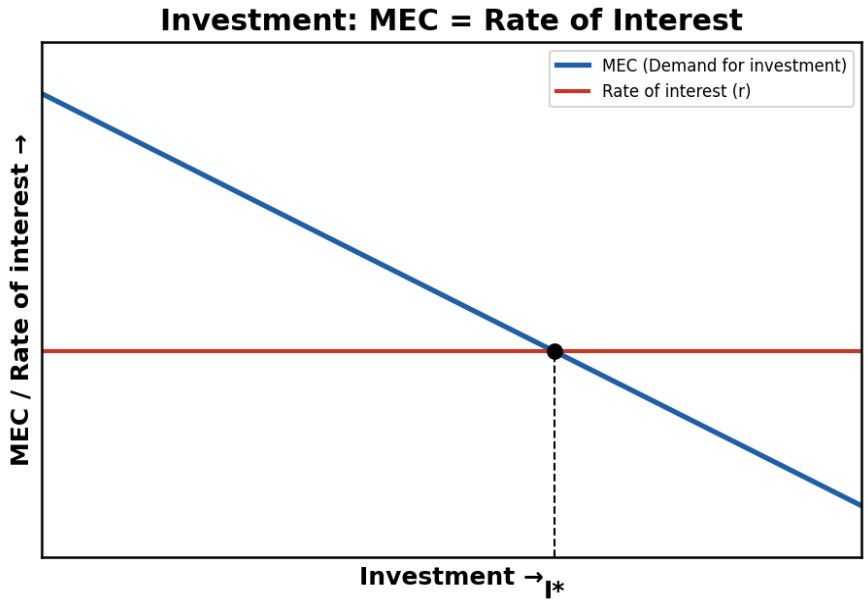


Fig 6.2 — Investment equilibrium jahan MEC = rate of interest; r girne par investment I badhta hai.*

6.3 Accelerator Principle

Accelerator batata hai ki **induced investment** output/consumption mein **change (rate of change)** par depend karta hai, level par nahi. Agar demand badh rahi hai to firms naya capital (machines) lagati hain.

$$I_t = v\,(Y_t - Y_{t-1}) \quad [v = \text{accelerator / capital-output ratio}]$$

Super-multiplier (multiplier + accelerator interaction): Samuelson-Hicks ne dikhaya ki multiplier aur accelerator milke ek doosre ko reinforce karte hain — isse **business cycles** (ups & downs) generate hote hain.

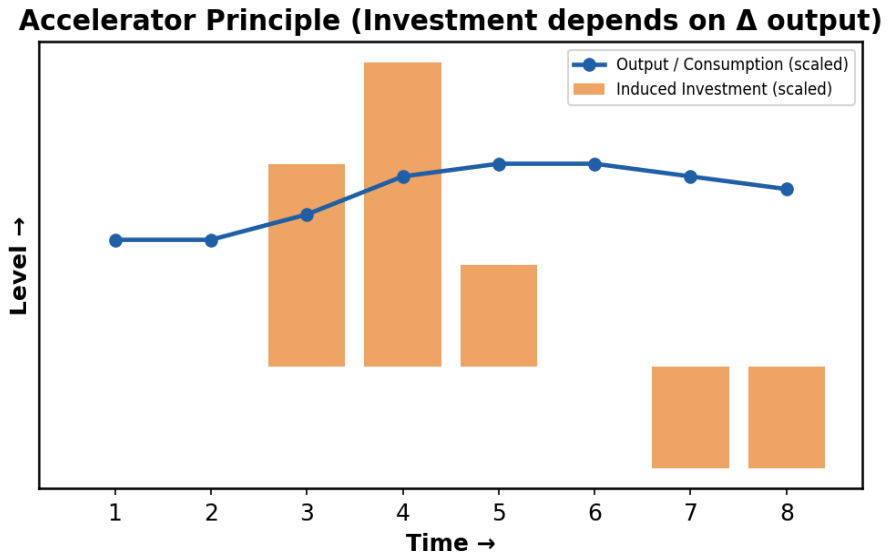


Fig 6.3 — Accelerator: output mein change badhne par induced investment tezi se badhta hai (aur change rukne par gir jaata hai).

6.4 Output–Price Determination in a Four-Sector Economy

Four-sector economy mein **Households, Firms, Government, aur Foreign sector** hote hain. Aggregate Demand = **C + I + G + (X − M)**. Equilibrium output aur price level **AD aur AS** ke intersection par. Government spending (G) ya net exports (X–M) badhne par AD curve **right** shift → output aur price level dono badhte hain (multiplier effect ke saath).

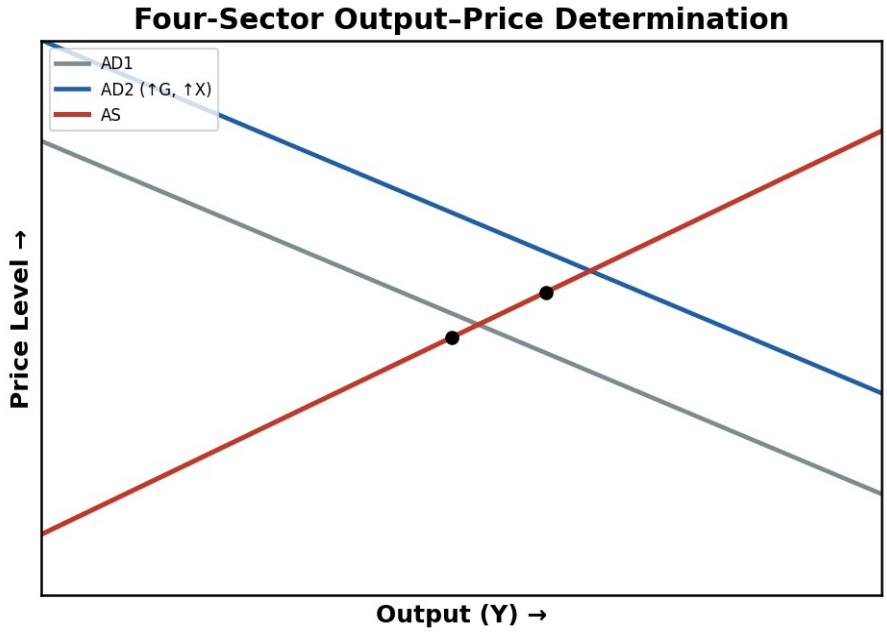


Fig 6.4 — Four-sector: G ya (X–M) badhne par AD right shift → naya higher equilibrium output & price.

7. IS–LM, Liquidity Preference, Mundell–Fleming & Trade Cycles

7.1 Liquidity Preference Theory of Interest (Keynes)

Keynes ke according interest ek **monetary phenomenon** hai — ye money ki demand aur supply se tay hota hai, savings se nahi. **Liquidity Preference** = log cash (liquid form) mein paisa rakhna kyun chahte hain. Teen motives:

- **Transaction Motive:** rozmarra ke transactions ke liye cash (income par depend).
- **Precautionary Motive:** unexpected/emergency needs ke liye cash.
- **Speculative Motive:** bonds vs cash ka choice — interest rate par depend (r↑ → speculative demand↓).

Interest rate wahan tay hota hai jahan **money demand (L) = money supply (M)**. Bahut low interest par **Liquidity Trap** aata hai — log saara extra money cash mein hold kar lete hain, interest aur neeche nahi girta, aur monetary policy ineffective ho jaati hai.

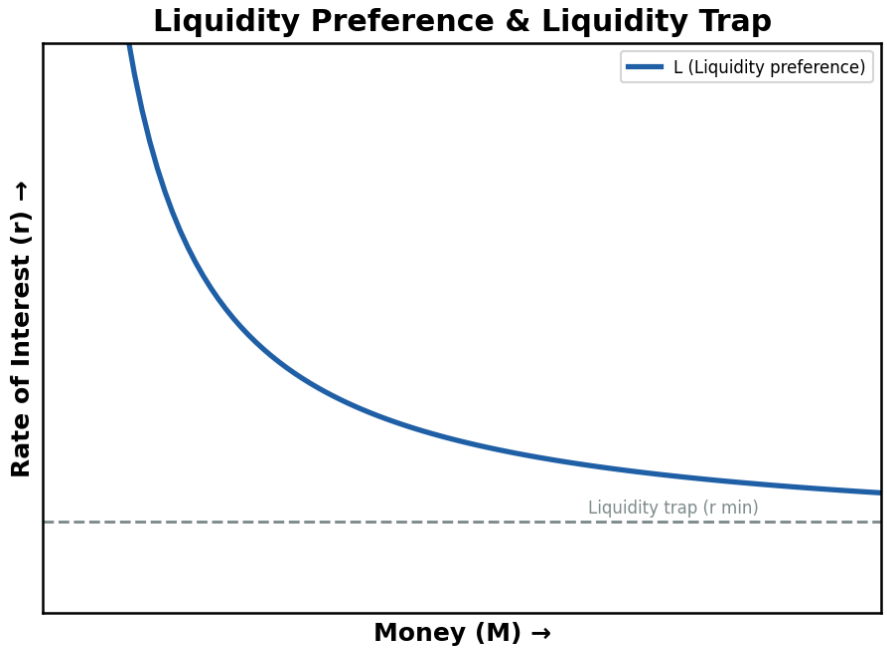


Fig 7.1 — Liquidity preference curve; bahut low r par horizontal portion = liquidity trap.

7.2 IS–LM Model (Hicks–Hansen)

IS–LM model Keynesian theory ko general-equilibrium roop mein dikhata hai — ye **goods market** aur **money market** dono ka simultaneous equilibrium ek saath (income Y aur interest r) determine karta hai.

- **IS Curve:** goods market equilibrium (Investment = Saving) ke saare combinations of Y aur r. **Downward sloping** — r↓ → I↑ → Y↑.
- **LM Curve:** money market equilibrium (Money demand L = Money supply M) ke combinations. **Upward sloping** — Y↑ → money demand↑ → r↑.

Dono curves ka intersection (point E) **simultaneous equilibrium** deta hai — equilibrium income Y* aur equilibrium interest rate r*. Fiscal policy (G, T) IS curve shift karti hai; monetary policy (M) LM curve shift karti hai.

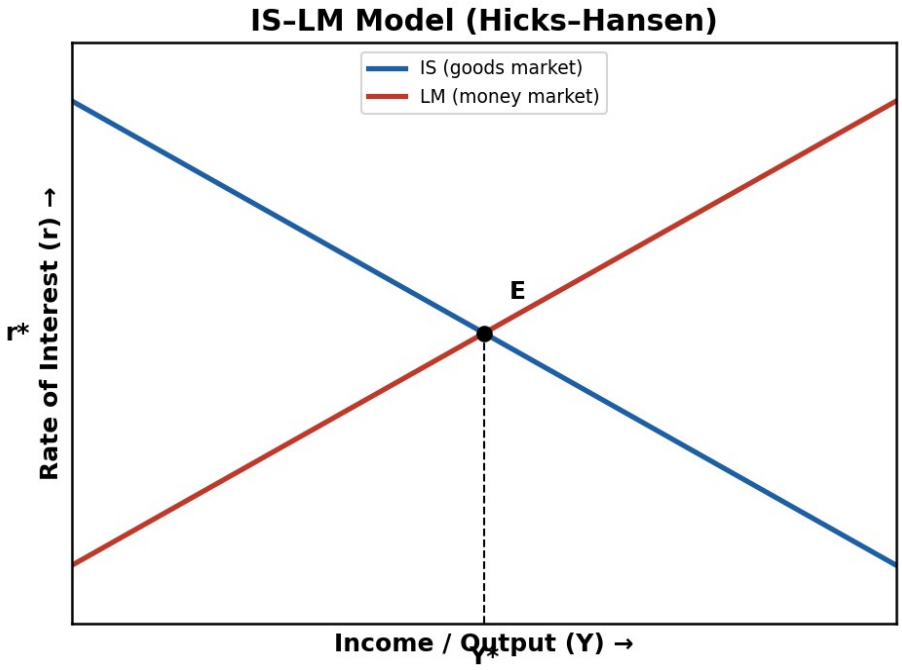


Fig 7.2 — IS (goods market) aur LM (money market) ka intersection E: Y aur r* ek saath determine.*

IS–LM Policy Effects

Expansionary Fiscal (G↑): IS right shift → Y↑, r↑ (crowding out possible). **Expansionary Monetary (M↑):** LM right shift → Y↑, r↓. Liquidity trap mein LM flat hone se monetary policy beasar, fiscal policy zyada effective.

7.3 Fleming–Mundell Open Economy Model

Mundell–Fleming model IS–LM ko **open economy** ke liye extend karta hai — yani international trade aur capital flows ko include karta hai. Ye IS–LM mein ek teesra curve **BP (Balance of Payments) curve** jodta hai jo external balance dikhata hai.

Central conclusion (Impossible Trinity / Trilemma): ek desh ek saath teeno nahi rakh sakta — (i) free capital movement, (ii) fixed exchange rate, aur (iii) independent monetary policy. Sirf 2 choose kar sakte hain.

- **Fixed exchange rate + free capital flows:** monetary policy ineffective, fiscal policy effective.
- **Flexible exchange rate + free capital flows:** monetary policy effective, fiscal policy crowding-out se kamzor.

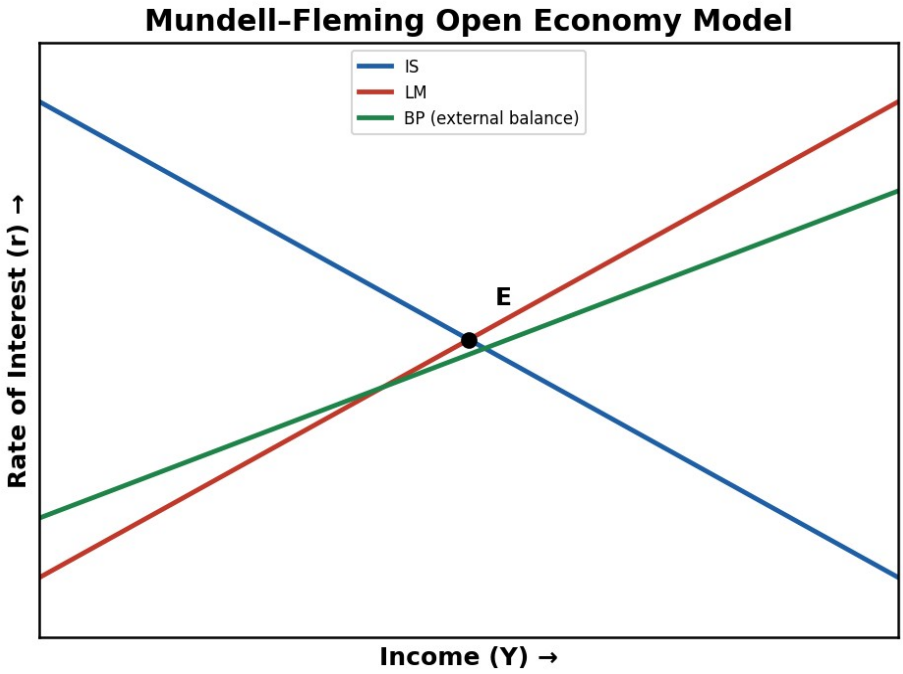


Fig 7.3 — Mundell–Fleming: IS, LM aur BP curves ka common point internal + external equilibrium.

7.4 Phillips Curve (Inflation–Unemployment)

A.W. Phillips ne dikhaya ki **inflation aur unemployment** mein short-run mein **inverse (trade-off)** relationship hai — low unemployment ke liye higher inflation tolerate karni padti hai. Lekin **Friedman & Phelps** ne kaha long-run mein ye trade-off nahi rehta (expectations adjust ho jaati hain) → long-run Phillips curve **vertical** at 'natural rate of unemployment'.

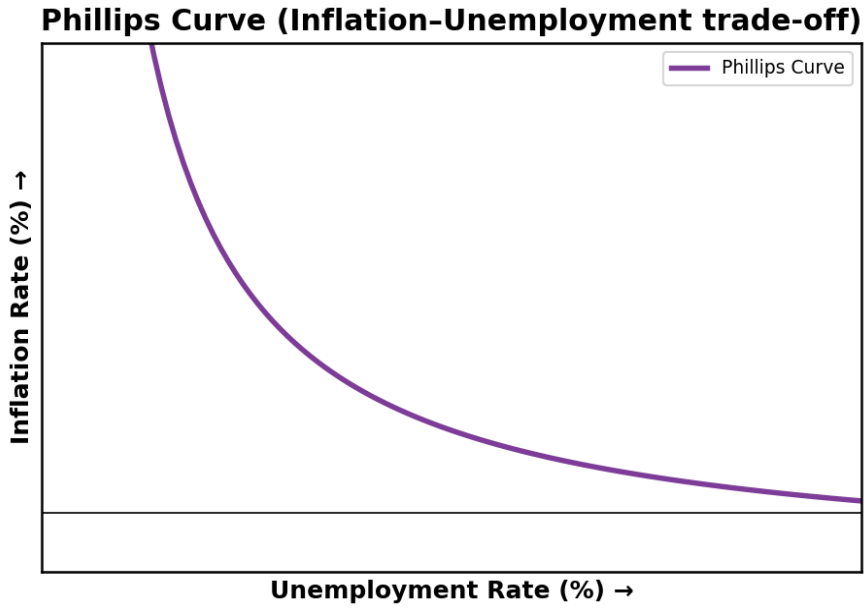


Fig 7.4 — Short-run Phillips curve: inflation ↑ → unemployment ↓ (trade-off).

7.5 Trade / Business Cycle Theories

Trade Cycle (Business Cycle) = economic activity (output, employment, income) mein recurring ups & downs. Iski **4 phases** hoti hain:

1. **Boom / Peak:** high output, employment, prices; full capacity, optimism.
2. **Recession:** activity girna shuru — demand ↓, investment ↓, unemployment ↑.
3. **Depression / Trough:** sabse neeche — low output & employment, pessimism (e.g. Great Depression 1929).
4. **Recovery / Expansion:** demand & investment phir badhna — economy upar ki taraf.

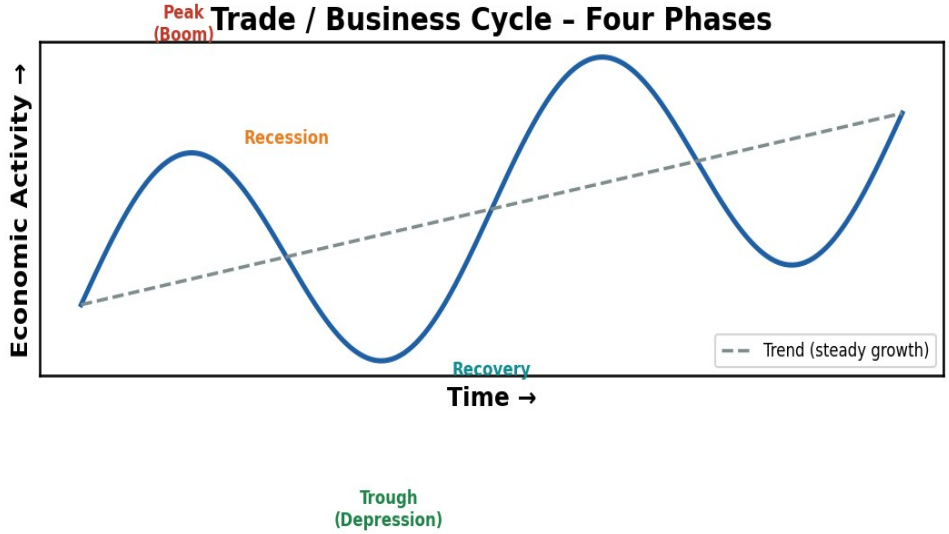


Fig 7.5 — Business cycle ki char phases trend line ke around: Boom, Recession, Depression, Recovery.

Important Theories of Trade Cycle

- **Hawtrey – Monetary Theory:** cycles mainly money/credit ki supply ke fluctuations se aate hain.
- **Keynes:** cycles ki jad — MEC (investment expectations) ke fluctuations aur effective demand mein changes.
- **Hicks & Samuelson – Multiplier-Accelerator:** multiplier aur accelerator ka interaction self-generating cycles paida karta hai (upper & lower limits ke beech).
- **Schumpeter – Innovation Theory:** innovations ke 'swarms' se boom, aur unke khatam hone par recession.
- **Hayek – Over-investment / Monetary Over-investment:** artificially low interest se zyada investment → eventual bust.

> **Quick Recall — Key Macro Formulae**
>
> **GNP = GDP + NFIA; NI = NNP at Factor Cost. Y = C + I + G + (X−M). C = a + bY; MPC + MPS = 1. Multiplier k = 1/(1−MPC) = 1/MPS. Accelerator It = v(Yt − Yt−1). Investment equilibrium: MEC = r.**

— End of Unit 2 : Macro-Economic Analysis —